# Physics-Informed Neural Networks and Neural-Mechanistic Hybrids for a WNT–RA–HOX Model of Colorectal Cancer Stemness

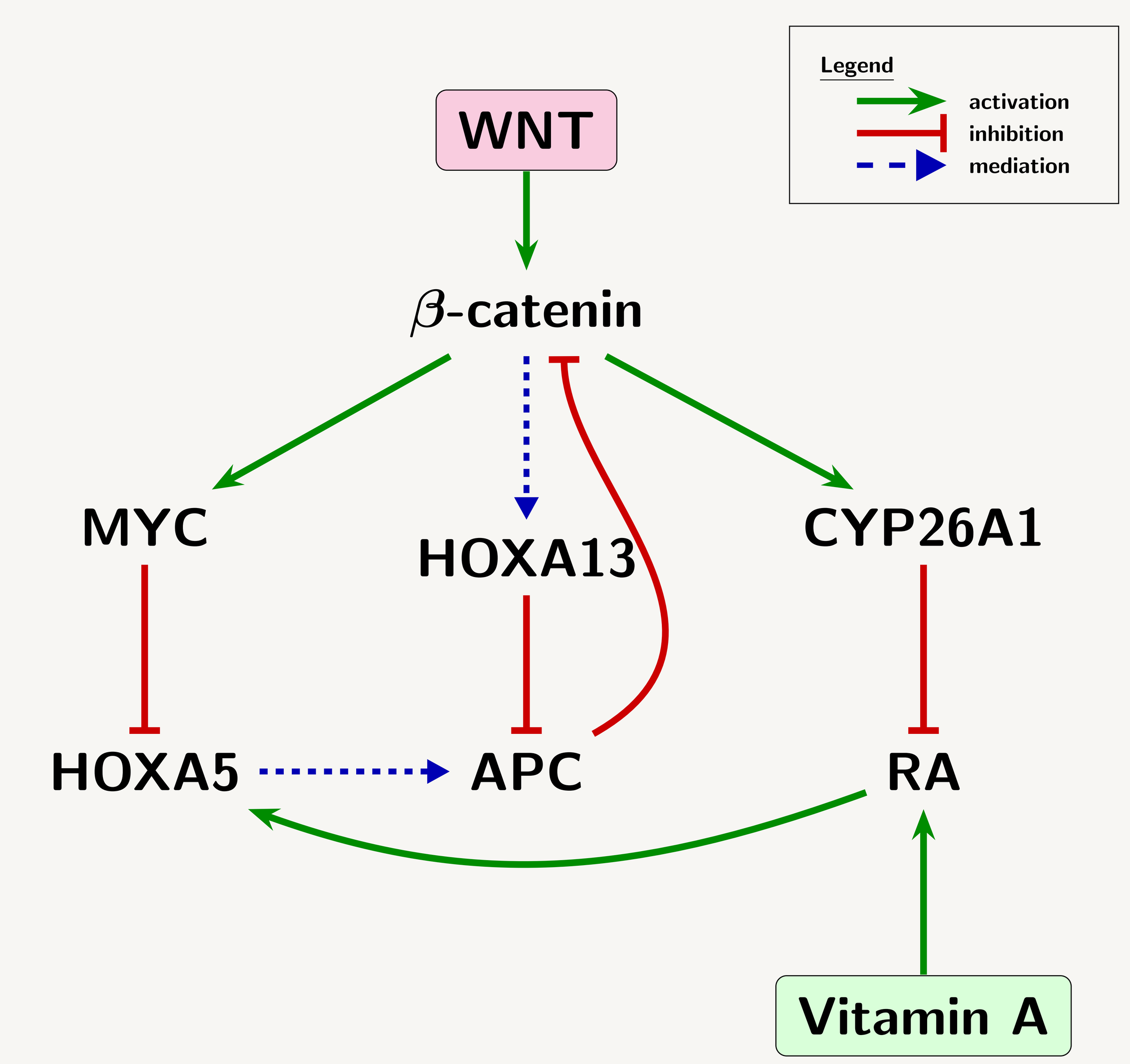
Sparse data PINNs · identifiability · experiment design

Amirkhan Aidarkhan · Pascal Kataboh

## 1 WNT–RA–HOX in seven states.


**Abstract**

APC loss deregulates WNT/β-catenin signalling; retinoic acid counters it through HOX. We study a nondimensional seven-state ODE system across four disease regimes. A Fourier-feature PINN solves it from 40 observations; an integral residual improves inverse recovery. Fisher and posterior analyses expose the remaining high-WNT ambiguity. In a neural-mechanistic hybrid, one targeted siRNA restores identifiability by making the data visit the learned term's zero anchor.


Green activates, red inhibits, blue mediates.

### Dimensionless system

$$\boldsymbol{y} = [b, p, h_5, h_{13}, m, r, c]^T, \quad X = X_0 x, \quad \tau = d_B t$$

$$\varepsilon_x \dot{x} = g_{x(\boldsymbol{y};\theta)} - x, \quad \theta \in \mathbb{R}^{36}$$

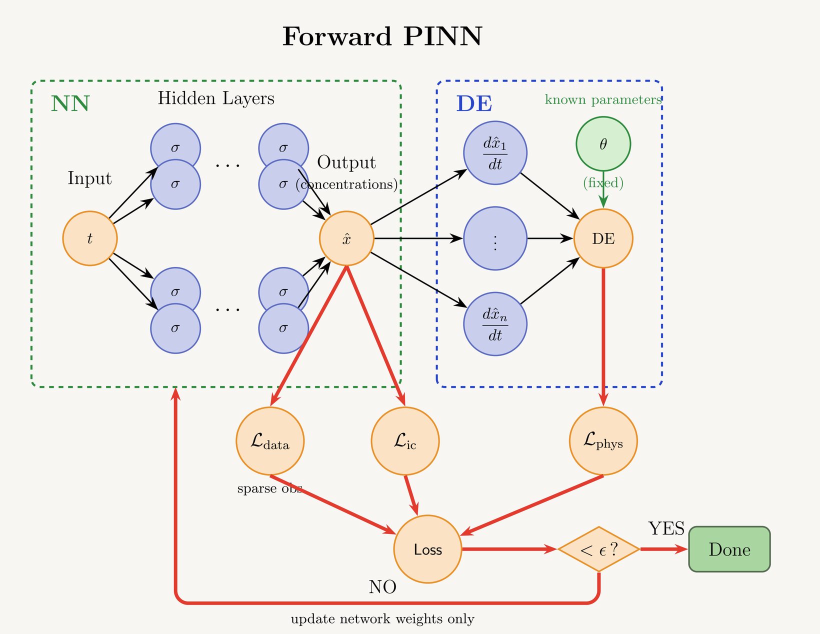
$$\dot{b} = W + \eta_{13} \frac{h_{13}^n}{\kappa_{13}^n + h_{13}^n} - b - \lambda_P p b - \lambda_5 \frac{h_5 b}{\kappa_5 + b}$$

$$\varepsilon_5 \dot{h}_5 = a_5 + \eta_R \frac{r}{\kappa_R + r} - h_5 - \eta_M \frac{m h_5}{\kappa_M + m}$$

$$S(\tau) = \frac{b(1+\alpha_{13} h_{13})}{(1+p)(1+\alpha_5 h_5)}$$

Normal → Severe APC Loss: $W = 0.8 \to 2.0$; $\theta_P = 1.0 \to 0.25$.

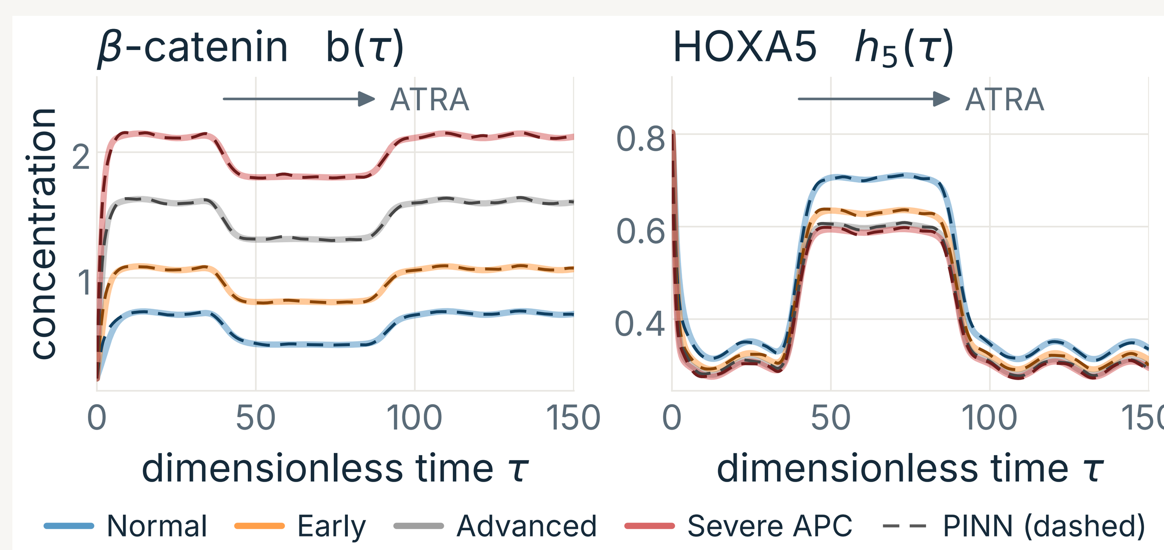
## 2 Sparse trajectories, then inversion.




Fourier features resolve fast forcing; θ is fixed forward and trainable inverse.

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \mathcal{L}_{\text{phys}} + 20\mathcal{L}_{\text{ic}}$$

$$\gamma(\tau) = \left[\tfrac{\tau}{T},\ \sin(2\pi \tfrac{\tau}{T} B_k),\ \cos(2\pi \tfrac{\tau}{T} B_k)\right]_{k=1}^{16}$$

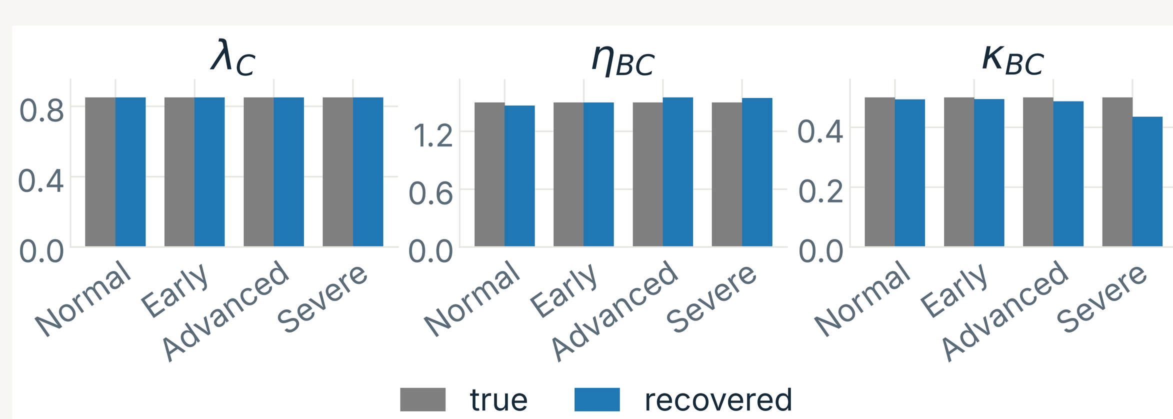



40 samples plus the ODE residual; PINN predictions are dashed.

| **1.06%** | **2.41%** |
|---|---|
| dense labels | 40 observations |

### Integral inverse residual

$$r_k = (z_{k+1} - z_k) - \frac{\Delta\tau_k}{2}(f_k + f_{k+1})$$



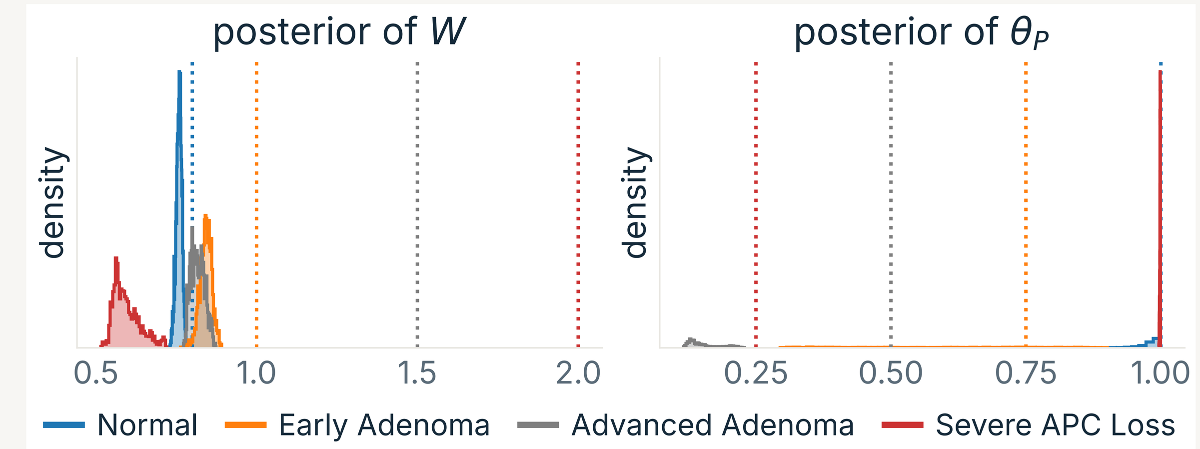

True (grey) and recovered (blue).

**Under 10% error**

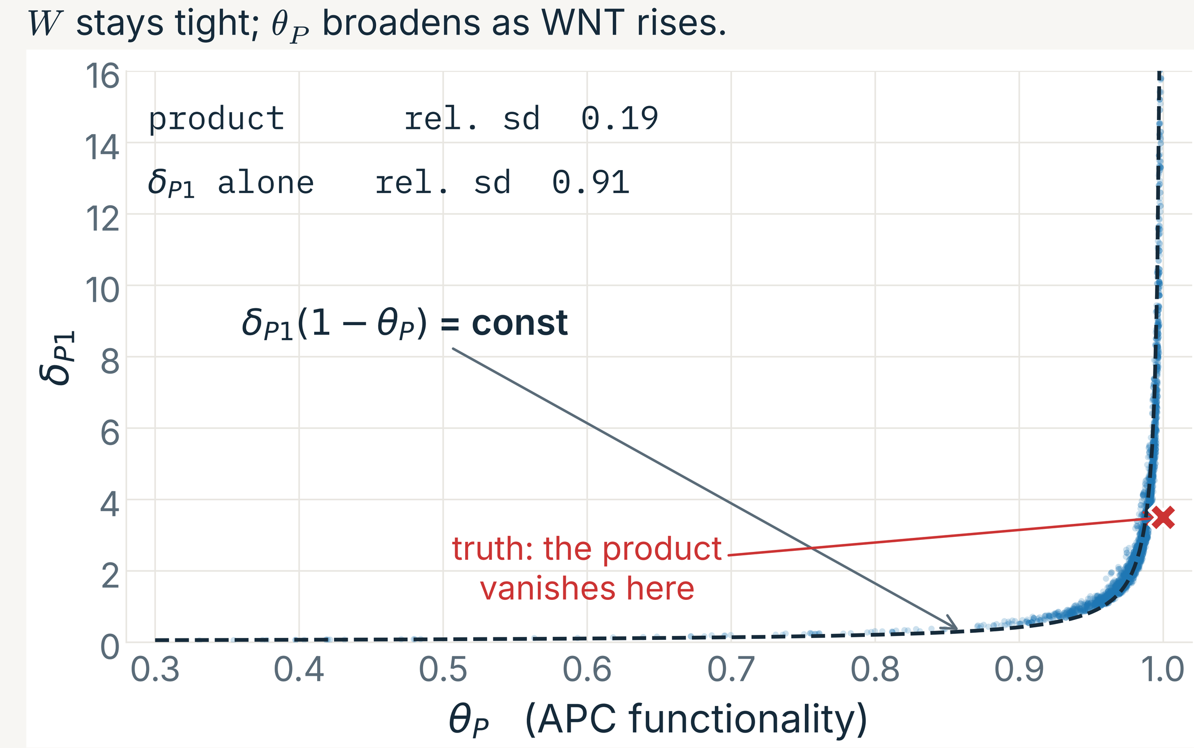
10/4/5/7 → 17/16/10/7 Across regimes: 37 → 50 of 144 parameters.

## 3 High WNT hides APC functionality.

$$p(\theta \mid \mathcal{D}) \propto \exp\left(-\frac{\mathcal{L}_{\text{int}(\theta)}}{2\sigma^2}\right) p(\theta)$$

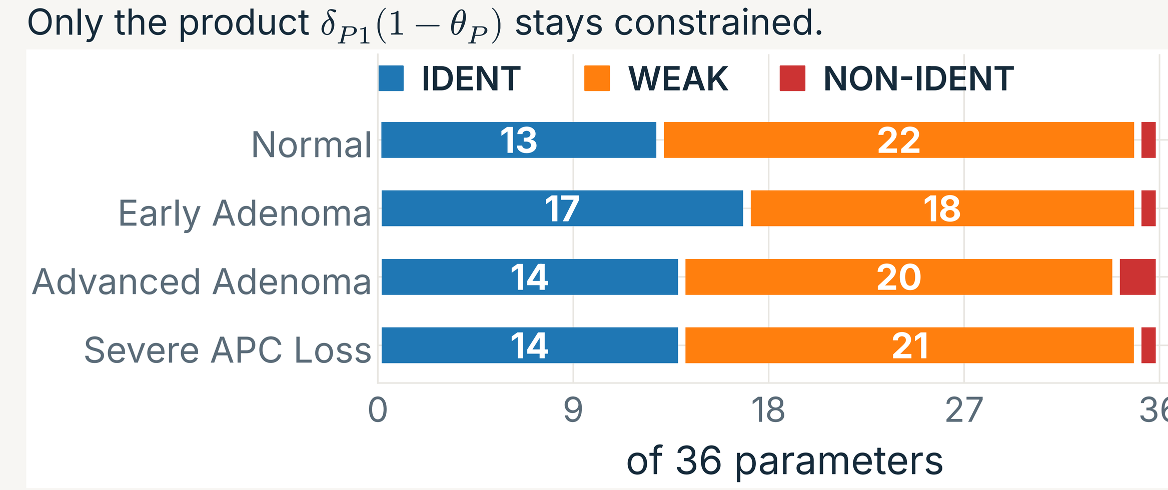



$W$ stays tight; $\theta_P$ broadens as WNT rises.




$$\delta_{P(\theta_P)} = 1 + \delta_{P1}(1 - \theta_P)$$

Only the product $\delta_{P1}(1-\theta_P)$ stays constrained.




Independent Fisher analysis gives the same verdict.

**Calibration check**

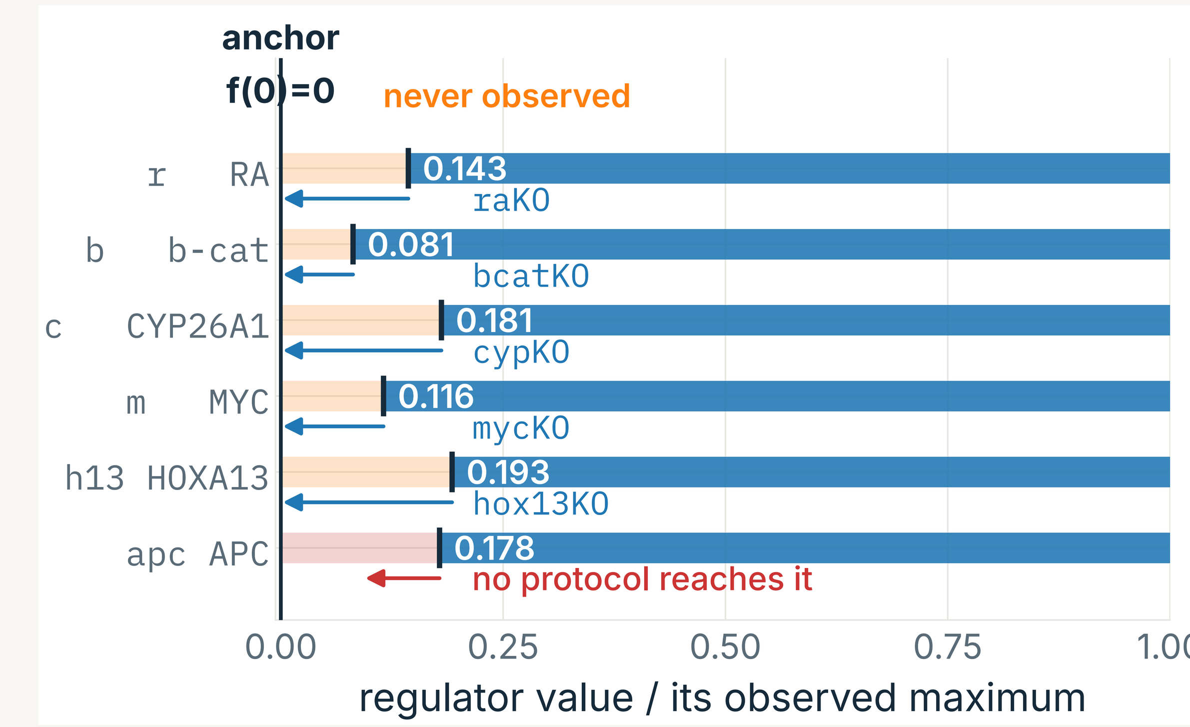
Recovery: **19/20/11/8**. Coverage: **13/36.** Interpret posterior geometry, not interval widths.

## 4 Visit the constraint.

$$\varepsilon \dot{x} = a + f_{\text{NN}}(u) - x, \quad f_{\text{NN}}(0) = 0$$

The zero anchor protects the basal parameter $a$.

Five parameterisations still leave $a$ 14 to 203% wrong.

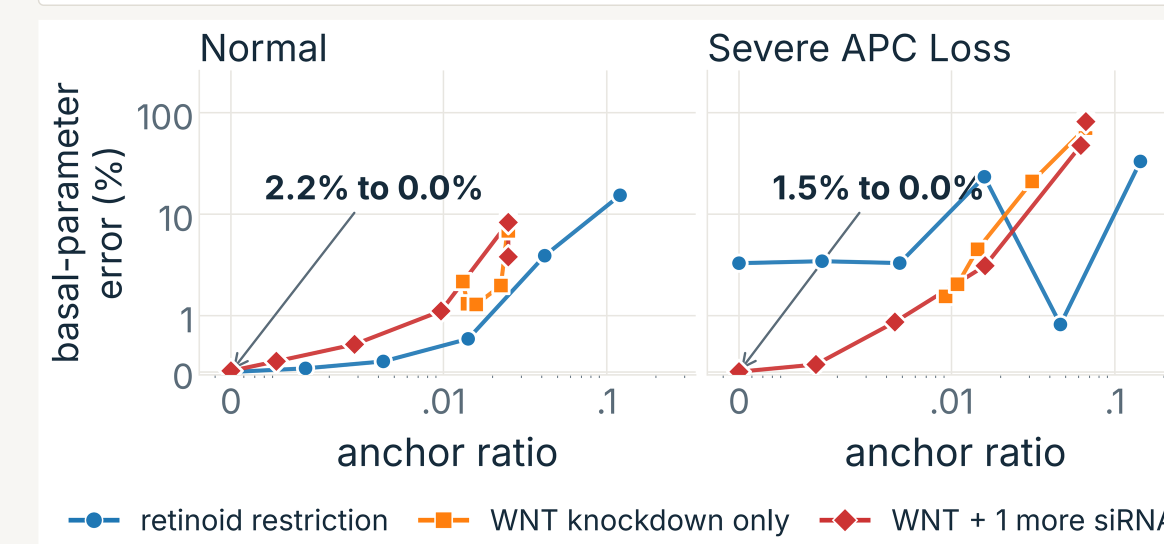



Ten conditions never approach the anchor.

### One targeted condition

| misses anchor | 1. one siRNA |
|---|---|
| **2.2% / 1.5%** | **0.0% / 0.0%** |

Normal / Severe APC Loss; eleven conditions per arm.




Error collapses by anchor ratio, not by dose.

**Experiment design restores identifiability.**


**Key references**

M. Raissi, P. Perdikaris, G. E. Karniadakis. Physics-informed neural networks. J. Comput. Phys. **378**, 686 (2019).

C. Rackauckas et al. Universal differential equations for scientific machine learning. arXiv:2001.04385 (2020).